

Week 1 — Foundations: arrays & DataFrames

- Lecture slides
- NotebookMind demo course
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Course intro & tooling

- Why NumPy and pandas
- The DataFrame mental model
- Reproducibility with seeds

NumPy arrays

- Vectorised operations vs. Python loops
- dtypes and shapes
- Random number generation

pandas DataFrame

- Building from a dict of columns
- `head()`, `info()`, `describe()`
- Selecting columns and rows

Reproducible data

- `np.random.seed`
- Why determinism matters
- Lab: build the students DataFrame